

What type of attack is this?

Brute Force Attack.

Which IP is suspicious?

The IP address 192.168.1.50 is suspicious.

Was the attack successful?

Yes the attack was successful.

What action should be taken?

The IP address 192.168.1.50 should be immediately blocked, the user1 account should be locked and its password reset, and a full investigation should be launched to find out what the attacker accessed after logging in.

Detailed Log Analysis

Looking at the log data the timeline of the attack is very clear:

At 10:17:45 the first failed login attempt was made for user1 from 192.168.1.50. At 10:18:02 the second failed attempt was made, just 17 seconds after the first. At 10:18:15 the third failed attempt followed 13 seconds later. At 10:18:30 the fourth failed attempt came 15 seconds after that. Finally at 10:20:10 a successful login was recorded from the exact same IP address.

The extremely short time gap between each attempt clearly proves that these were not manual login attempts by a human. A real human cannot try four different passwords within 45 seconds. This confirms that an automated brute force tool was used to rapidly try multiple password combinations.

2. Steps Taken by the SOC Team**L1 Analyst — Alert Detection and Triage**

The L1 analyst was the first to receive the SIEM alert about multiple failed login attempts on the server. They quickly reviewed the alert details and confirmed that four failed attempts followed by one successful login had all come from the same IP address 192.168.1.50 within a very short time period. The L1 analyst classified this as a high severity incident and immediately escalated it to the L2 analyst for deeper investigation without wasting time.

L2 Analyst — Investigation and Containment

The L2 analyst took the escalated alert and began a thorough investigation. They pulled the complete login logs for user1 and confirmed the brute force pattern. They then checked what

actions the attacker performed after the successful login at 10:20:10 to assess how much damage was done. After confirming the attack the L2 analyst took immediate containment actions by blocking the IP address 192.168.1.50 on the firewall, locking the user1 account to prevent further access, and preserving all log evidence for the investigation report.

L3 Expert — Rule Creation and Prevention

The L3 expert reviewed the full incident report submitted by L2 and took long term preventive action. They created a new detection rule in the SIEM system that would automatically trigger a critical alert and block any IP address that fails to login more than 3 times within 60 seconds. They also wrote a detailed incident report documenting the attack timeline, impact assessment, and all actions taken, and presented recommendations to management to prevent similar attacks in the future.

3. Prevention Methods

Enable Account Lockout Policy The system should be configured to automatically lock a user account after a certain number of failed login attempts such as 3 to 5 attempts. This would have locked the user1 account before the attacker could succeed even if they kept trying different passwords.

Use Strong and Complex Passwords The fact that the attacker succeeded after only four attempts suggests that user1 had a very weak password. All users should be required to use strong passwords that contain uppercase letters, lowercase letters, numbers, and special characters. Weak passwords are cracked very easily by brute force tools.

Enable Two Factor Authentication Even if an attacker correctly guesses the password, two factor authentication would stop them from logging in because they would also need a second verification such as a one time code sent to the user's phone. This is one of the most effective ways to prevent unauthorized access.

Regular Log Monitoring The security team should regularly monitor authentication logs and set up SIEM rules to detect brute force patterns early. In this case the attack could have been stopped at the first few failed attempts if proper monitoring rules were in place.